

When Neural Models Can Compose but Cannot Diagnose: The Decomposition Gap in Predictive State Revision

First Author¹, Second Author¹, and Third Author²

¹ Institution One, City, Country
`first.author@email.com`

² Institution Two, City, Country
`third.author@email.com`

Abstract. Neural models that reason about dynamic environments must revise the correct parts of an internal predictive state when multiple causes of change co-occur. Failures in compositional state revision can arise from two sources: an inability to compose known updates, or an earlier inability to identify which causes are active in a mixed observation. We disentangle these in a controlled predictive-map benchmark where before/after observations reflect changes in sensory appearance, value structure, topology, and action effects. The benchmark compares direct inference from a mixed change against an oracle tuple setting in which the same change is decomposed into its single-cause components. Under leave-one-composition-out evaluation, standard CNNs, shared-gate models, and independent gate-head models solve seen compositions but fail on heldout mixed changes, with the best model reaching only 19.7% exact match. When the same heldout cases are supplied with oracle single-cause decompositions, tuple-based composition reaches 99.8–100.0% exact match, yielding direct-vs-oracle gaps of 95.6–100.0 percentage points. Seen-pair controls reach ≈ 99.9 –100.0%, and strict dataset audits confirm balanced labels and meaningful component evidence across folds. The bottleneck is not update composition but cause decomposition: the tested models can use separated causes but do not reliably infer them from raw mixed observations.

Keywords: Neural network models · Compositional generalization · Predictive state representations · Causal representation learning

1 Introduction

Neural models are increasingly used as predictive systems for environments that change over time. A model must not only predict future observations but also revise the right parts of its internal state when the world changes—a requirement central to world models and model-based agents [4, 6, 5]. A model that updates

the wrong component may preserve short-term accuracy in familiar cases while failing when the same causes appear in a new configuration.

The difficulty becomes sharper when changes are mixed. A single before/after observation may reflect several simultaneous causes: an object looks different, a goal moves, a path becomes blocked, or an action produces a different effect. A model can appear competent if it has memorised the same combination during training, but this does not imply it has learned reusable, cause-specific updates. Compositional generalization work has documented this gap repeatedly: neural models achieve high in-distribution accuracy while failing on systematic recombinations of familiar parts [10, 8, 9, 15]. However, a failed heldout combination does not identify *why* it fails.

This paper focuses on a specific ambiguity. When a model fails on an unseen mixture of changes, the failure may arise because it cannot *compose* the corresponding updates once the causes are known, or earlier, because it cannot *decompose* the mixed observation into the individual active causes. These two failure modes point to different solutions: if composition is the bottleneck, stronger modular update operators may help; if decomposition is the bottleneck, better gates alone will not suffice.

We study this distinction in a controlled predictive-map benchmark evaluated under a leave-one-composition-out protocol. We compare a *direct* setting, in which the model must infer active causes from mixed observations, against an *oracle tuple* setting, in which the same heldout case is supplied with its separated single-cause components. This direct-vs-oracle comparison reveals a large decomposition gap: all direct models fail on heldout mixtures while oracle tuple composition nearly solves the same cases when decomposition is provided. Seen-pair controls confirm learnable tasks; dataset audits rule out labelling artifacts. Together, the results isolate cause decomposition—not update composition—as the primary bottleneck in the tested setting.

2 Related Work

Compositional generalization. Systematic evaluation of compositional generalization spans multiple domains. SCAN [10] showed that sequence-to-sequence models fail on systematic recombinations of known primitives. CFQ [8] and COGS [9] extend this to question answering and semantic interpretation. CLEVR [7], gSCAN, and ReaSCAN [15, 17] provide visually grounded diagnostics. Our work follows this tradition but shifts focus to predictive-state revision, and asks whether failure arises from update composition or from the earlier decomposition of mixed evidence—a distinction prior benchmarks do not address.

World models and predictive state. World models learn representations supporting prediction and control [4, 6, 5]. Predictive state representations characterise state via predictions about future observations rather than explicit latent variables [11]. Our benchmark isolates a failure mode relevant to any learned predictive state: when multiple aspects of the environment change together, can a model revise the correct state components?



Fig. 1. Diagnostic logic. The same heldout composed change is evaluated under three conditions. Direct inference fails; oracle tuple composition and seen-pair controls both succeed. This pattern isolates cause decomposition as the bottleneck.

Modularity and reusable mechanisms. Neural module networks [1], Recurrent Independent Mechanisms [3], Slot Attention [12], and independent causal mechanisms [14, 16] all argue that factorised structure improves generalisation. Our results are consistent with this motivation but identify a prior requirement: before modular updates can be applied correctly, the model must infer which mechanisms are active in the mixed observation.

Causal diagnostics. Causal abstraction [2] and activation patching [13] test whether model components play a causal role in predictions. In our setting, gate patching is confounded when routing gates also serve as label logits; a decoupled analysis does not yield reliable state repair. We treat patching as a diagnostic limitation (Section 6).

3 Benchmark and Evaluation Protocol

We study compositional state revision in a controlled predictive-map environment. Each example is a before–after pair of grid observations; the model predicts a four-dimensional multi-label target $y \in \{0, 1\}^4$ indicating which cause types are active. We consider four cause types—sensory appearance, value structure, topology, and action effects—each affecting a qualitatively distinct aspect of a predictive state. Single-cause examples have one active entry; composed examples have two. We use exact match as the primary metric:

$$\text{Exact}(\hat{y}, y) = \mathbf{1}[\hat{y}_1 = y_1, \hat{y}_2 = y_2, \hat{y}_3 = y_3, \hat{y}_4 = y_4]. \quad (1)$$

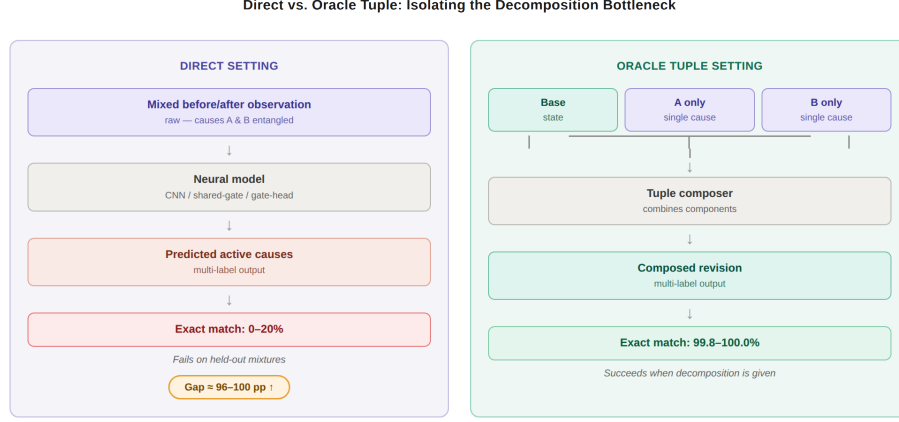


Fig. 2. Direct mixed inference versus oracle tuple composition. In the direct setting the model observes only a mixed before–after change. In the oracle tuple setting the same case is provided with its single-cause components, separating composition from cause decomposition.

A prediction is correct only when every active and inactive cause is identified correctly.

3.1 Direct and Oracle Tuple Settings

In the **direct** setting the model receives only the mixed before–after observation and must infer all active causes. In the **oracle tuple** setting the same composed change is provided together with its single-cause components (base, *A*-only, *B*-only, and *A+B*). The oracle tuple is not a fair competing baseline—it receives privileged information unavailable at test time—but serves as a diagnostic upper bound on composition ability.

We define the *decomposition gap* as

$$\Delta_{\text{decomp}} = E_{\text{oracle}} - E_{\text{direct}}, \quad (2)$$

where E_{oracle} and E_{direct} are oracle tuple and direct exact match on the same heldout fold. A large gap with poor direct but near-perfect oracle performance indicates that composition is not the bottleneck.

3.2 Leave-One-Composition-Out Protocol

The model is trained on all single-cause examples and a subset of composed pairs; one composed pair is held out for evaluation. We construct four folds: value+topology, sensory+action, value+action, and sensory+topology (Figure 3). The heldout pair never appears as a composed training example, but both constituent causes appear individually and in other composed pairs, ensuring failure cannot be attributed to unfamiliarity with individual causes.

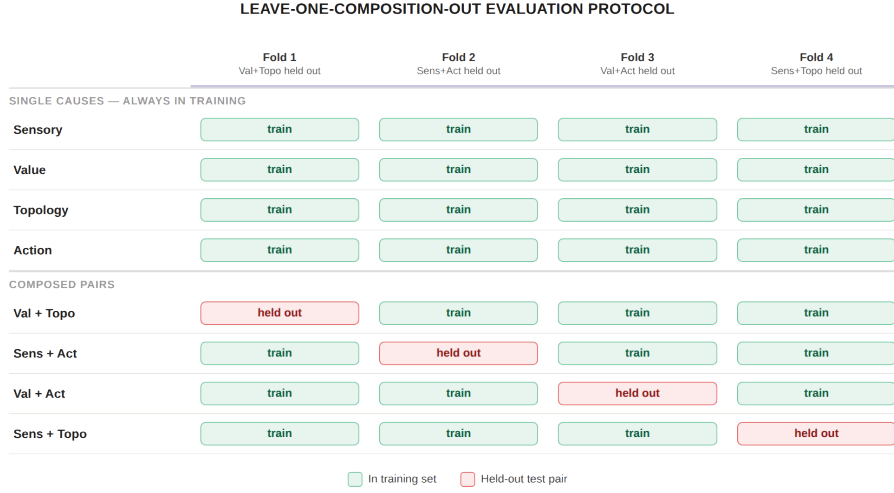


Fig. 3. Leave-one-composition-out protocol. Single-cause examples appear in every training split. One composed pair is held out per fold; remaining composed pairs are in training.

3.3 Controls and Audits

Seen-pair upper-bound controls train models with all composed pairs included; high performance confirms learnable tasks. *Dataset audits* verify class balance and component evidence, ensuring poor heldout performance is not explained by missing signal, invalid labels, or task impossibility.

4 Models and Experimental Setup

We evaluate three direct-inference models and one oracle diagnostic, testing whether common neural designs can identify and recombine causes under the leave-one-composition-out protocol.

Standard CNN. A convolutional network receiving the concatenated before–after observation and predicting the multi-label target directly. No explicit pathway structure; tests whether generic supervised feature learning solves heldout mixtures.

Shared-gate predictive model. Encodes before/after observations into latent spatial states, computes error features, and predicts four gates g_k weighting pathway-specific updates. The revised latent state is $z' = z + \sum_{k=1}^4 g_k u_k$, where u_k is the pathway update for cause k .

EXPERIMENTAL DESIGN MATRIX			
CONDITION	INPUT PROVIDED	QUESTION TESTED	RESULT
Direct held-out leave-one-composition-out	Mixed before/after observation only obs(A+B) — causes entangled	Can the model infer which causes are active from mixed raw evidence?	✗ Fails 0–20% exact match best model: 19.7% avg.
Oracle tuple diagnostic upper bound	Base + single-cause components + composed base, A-only, B-only, A+B	Can updates compose correctly once causes are already separated?	✓ Succeeds 99.8–100.0% exact gap vs. direct: 95.6–100 pp
Seen-pair control upper-bound control	Held-out pair included in training A+B ∈ train set	Are held-out compositions learnable under direct supervision?	✓ Succeeds 99.9–100.0% exact rules out task impossibility
Dataset & evidence audit validity check	Generated labels + intervention evidence all folds, all cause types	Are labels valid, balanced, and component evidence present?	✓ All pass Pass rate: 1.000 rules out label artifacts

Fig. 4. Experimental design matrix summarising conditions, inputs, questions tested, and outcomes.

Independent gate-head model. Separate prediction heads per cause, shared evidence features. Independent heads reduce output-level interference but do not factorise the evidence representations into cause-stable features.

Oracle tuple composer. Receives the decomposed tuple (base, each single-cause component, composed change). Used only as a diagnostic upper bound; not a fair inference baseline.

All direct models are trained for 30 epochs on single-cause examples and seen composed pairs per fold; two seeds per model per fold (24 direct runs total). Model selection uses validation performance on seen compositions only. The oracle diagnostic runs once per fold. Seen-pair controls follow the same procedure with all composed pairs included.

5 Results

Figure 5 summarises all three conditions across the four heldout folds. Direct inference fails in every fold while oracle tuple composition and seen-pair controls reach near ceiling. This rules out two explanations: the compositions are learnable (seen-pair controls succeed) and updates are composable once separated (oracle tuple succeeds). The difficulty is inferring active causes from mixed observations.

5.1 Direct Models Fail on Heldout Mixtures

Table 1 reports average heldout exact match over folds and seeds. The standard CNN reaches 19.7%; the shared-gate and independent gate-head models reach 10.2% and 8.7%—both *worse* than the plain CNN, suggesting explicit gates fit training-set correlations in ways that hurt generalisation. Figure 6 shows consistent low scores across all four folds.

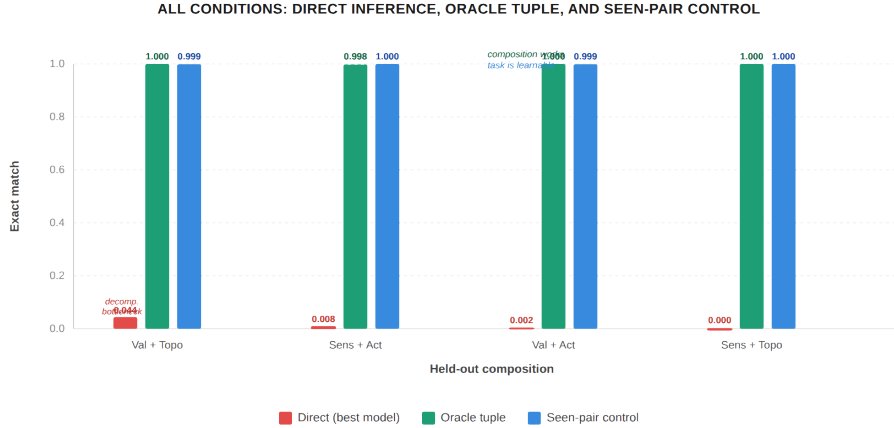


Fig. 5. All conditions across the four heldout folds. Direct inference fails while oracle tuple composition and seen-pair controls achieve near-perfect exact match, supporting the decomposition-gap interpretation.

Table 1. Direct heldout exact match (mean over four folds and two seeds).

Model	Heldout exact match	Interpretation
Standard CNN	0.197	Best direct baseline
Shared-gate model	0.102	Gates do not solve decomposition
Independent gate heads	0.087	Separate heads do not solve decomposition

5.2 Oracle Tuple Composition Nearly Solves the Same Cases

Table 2 shows the direct-vs-oracle gap per fold. Oracle tuple composition reaches 99.8–100.0% exact match; direct performance remains near zero. Gaps range from 95.6 to 100.0 percentage points. The heldout cases are not intrinsically hard to combine—once causes are supplied separately, composition nearly solves the task in every fold.

5.3 Controls Rule Out Alternative Explanations

Seen-pair controls reach ≈ 99.9 –100.0% exact match, confirming heldout compositions are learnable under direct supervision. All audits pass (Table 3): component evidence pass rate is 1.000 for every fold, and action-effect evidence is localised on or near the relevant path in all audited cases.

6 Analysis and Limitations

The failure is specific to unseen mixtures. All direct models are trained on single-cause examples; seen-pair controls confirm that composed cases are learnable

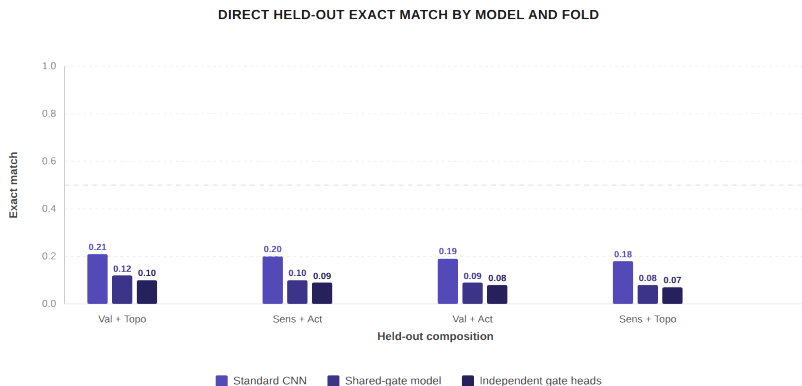


Fig. 6. Direct heldout exact match by model and fold.

Table 2. Direct-vs-oracle decomposition gap per heldout fold.

Heldout composition	Direct	Oracle	Gap
Value + Topology	0.044	1.000	0.956
Sensory + Action	0.008	0.998	0.990
Value + Action	0.002	1.000	0.998
Sensory + Topology	0.000	1.000	1.000

under direct supervision. The failure is specific to recognising familiar causes inside a heldout mixture never directly supervised—consistent with models that learn representations tied to observed training contexts rather than cause-stable features. The oracle tuple diagnostic rules out the interpretation that update operators cannot be combined; the seen-pair and audit controls rule out invalid or missing evidence. The remaining explanation—direct models cannot recover active causes from mixed observations—is positively supported by the decomposition gap and negatively supported by the elimination of all alternatives.

Explicit gating does not solve decomposition. The shared-gate and independent gate-head models underperform the plain CNN. This is not evidence against modularity in general, but against the assumption that adding gates or separate output heads suffices without a corresponding mechanism for forming cause-stable evidence. Gate activations appear to function as supervised labels for familiar contexts rather than reusable causal variables.

Causal patching does not localise the failure. Initial patching was confounded because routing gates also served as classification logits: gate-only, gate-plus-update, and gate-plus-zero-update interventions all produced nearly identical label rescue rates, indicating a gate-label coupling artefact rather than selective state repair. A decoupled gated model (routing gates separated from classification logits) passed the seen-pair sanity check at 99.9% but still failed on heldout

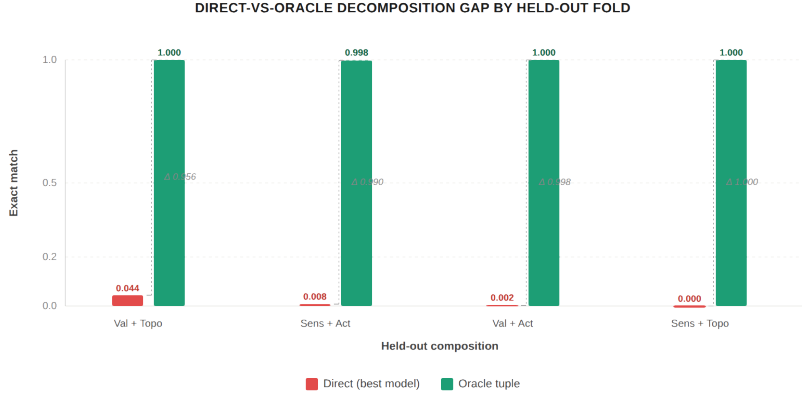


Fig. 7. Direct-vs-oracle decomposition gap. Direct inference is near floor; oracle tuple composition is near ceiling.

Table 3. Controls ruling out alternative explanations.

Control	Result
Seen-pair upper bound	0.999–1.000 exact match
Strict dataset audits	Passed for all folds
Strict tuple audits	Passed for all tuple-bearing folds
Component evidence pass rate	1.000 for all folds
Action evidence localisation	On-path or near-path rate 1.000

compositions (2.3% direct exact match), and decoupled patching still did not yield reliable state repair. Figure 8 and Table 4 summarise these diagnostics, treated as a limitation rather than a positive finding.

Table 4. Patching diagnostics (treated as a limitation).

Diagnostic	Coupled gates	Decoupled gates
Gate-only label rescue	High	Near zero
Gate-label confound	Present	Removed
Seen-pair sanity	High	0.999 exact match
Heldout direct exact	Low	0.023 average
Matched update state repair	Not identifiable	Not reliable

Implications and limitations. Progress requires mechanisms that form cause-stable evidence under mixed inputs, so update composition operates on the correct components. Object-centric representations [12], amortised inference in

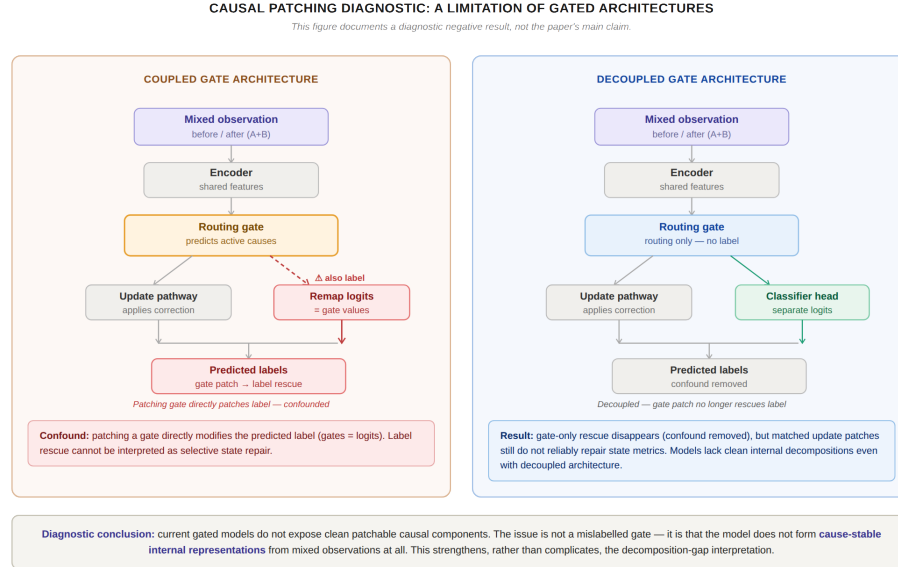


Fig. 8. Causal patching diagnostic. Coupled gates entangle routing with label prediction; decoupled gates remove this shortcut but matched update patching still does not produce selective state repair.

structured latent variable models, and contrastive objectives encouraging within-cause invariance are plausible directions. Limitations include the synthetic controlled setting (real-world gaps may differ), the oracle tuple being a diagnostic upper bound rather than a deployable method, and the restricted set of architectures tested.

7 Conclusion

We studied compositional state revision in a controlled predictive-map setting, asking whether neural models fail on heldout mixtures because they cannot compose known updates or because they cannot decompose mixed observations into active causes. Across four leave-one-composition-out folds, direct models failed consistently despite training on individual causes and other composed pairs; oracle tuple composition nearly solved the same cases when single-cause components were supplied; seen-pair controls and audits ruled out task impossibility, label artifacts, and absent evidence.

In this controlled setting the bottleneck is cause decomposition, not update composition. The tested models can use separated causes but cannot reliably recover them from raw mixed evidence. Stronger gates or modular pathways alone are unlikely to close this gap; progress requires mechanisms that form cause-stable evidence representations under mixed observations.

Acknowledgments. Acknowledgments to be added.

Disclosure of Interests. The authors have no competing interests to declare.

References

1. Andreas, J., Rohrbach, M., Darrell, T., Klein, D.: Neural module networks. In: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR). pp. 39–48 (2016)
2. Geiger, A., Wu, Z., Potts, C., Icard, T., Goodman, N.: Finding alignments between interpretability methods and formal notions of explanation. In: Uncertainty in Artificial Intelligence (UAI) (2023)
3. Goyal, A., Lamb, A., Hoffmann, J., Sodhani, S., Levine, S., Bengio, Y., Schölkopf, B.: Recurrent independent mechanisms. In: International Conference on Learning Representations (ICLR) (2021)
4. Ha, D., Schmidhuber, J.: World models. In: Advances in Neural Information Processing Systems (2018), also available as arXiv:1803.10122
5. Hafner, D., Lillicrap, T., Ba, J., Norouzi, M.: Dream to control: Learning behaviors by latent imagination. In: International Conference on Learning Representations (ICLR) (2020)
6. Hafner, D., Lillicrap, T., Fischer, I., Villegas, R., Ha, D., Lee, H., Davidson, J.: Learning latent dynamics for planning from pixels. In: Proceedings of the 36th International Conference on Machine Learning (ICML). pp. 2555–2565 (2019)
7. Johnson, J., Hariharan, B., van der Maaten, L., Fei-Fei, L., Zitnick, C.L., Girshick, R.: CLEVR: A diagnostic dataset for compositional language and elementary visual reasoning. In: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR). pp. 2901–2910 (2017)
8. Keyser, D., Schärli, N., Scales, N., Buisman, H., Sherburn, D., Puigcerver, J., Ruber, L., et al.: Measuring compositional generalization: A comprehensive method on realistic data. In: International Conference on Learning Representations (ICLR) (2020)
9. Kim, N., Linzen, T.: COGS: A compositional generalization challenge based on semantic interpretation. In: Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP). pp. 9087–9105 (2020)
10. Lake, B.M., Baroni, M.: Generalization without systematicity: On the compositional skills of sequence-to-sequence recurrent networks. In: Proceedings of the 35th International Conference on Machine Learning (ICML). pp. 2879–2888 (2018)
11. Littman, M.L., Sutton, R.S., Singh, S.: Predictive representations of state. In: Advances in Neural Information Processing Systems. vol. 14 (2001)
12. Locatello, F., Weissenborn, D., Unterthiner, T., Mahendran, A., Heigold, G., Uszkoreit, J., Dosovitskiy, A., Kipf, T.: Object-centric learning with slot attention. In: Advances in Neural Information Processing Systems. vol. 33, pp. 11525–11538 (2020)
13. Meng, K., Bau, D., Andonian, A., Belinkov, Y.: Locating and editing factual associations in GPT. In: Advances in Neural Information Processing Systems. vol. 35, pp. 17359–17372 (2022)
14. Parascandolo, G., Kilbertus, N., Rojas-Carulla, M., Schölkopf, B.: Learning independent causal mechanisms. In: Proceedings of the 35th International Conference on Machine Learning (ICML). pp. 4036–4044 (2018)

15. Ruis, L., Andreas, J., Baroni, M., Bouchacourt, D., Lake, B.M.: A benchmark for systematic generalization in grounded language understanding. In: Advances in Neural Information Processing Systems. vol. 33, pp. 19861–19872 (2020)
16. Schölkopf, B., Locatello, F., Bauer, S., Ke, N.R., Kalchbrenner, N., Goyal, A., Bengio, Y.: Toward causal representation learning. *Proceedings of the IEEE* **109**(5), 612–634 (2021)
17. Wu, Z., Kreiss, E., Ong, D.C., Potts, C.: ReaSCAN: Compositional reasoning in language grounding. In: Advances in Neural Information Processing Systems. vol. 34 (2021)